AMT-130: a historical prior from FDA advisory committee votes

4 November 2026

Summary

uniQure's AMT-130 is a gene therapy for Huntington's disease. The company has said that the U.S. Food and Drug Administration (FDA) accepted its biologics license application for accelerated approval, granted priority review, and set a target decision date of 1 May 2027, and that an advisory committee meeting is planned and not yet dated. Those statements come from uniQure releases on 2 and 4 November 2026. They are company statements until an FDA letter is in hand. The product's orphan-drug designation is an FDA record: designation 525216, dated 27 September 2017, for an adeno-associated virus serotype 5 therapy that delivers a microRNA aimed at the huntingtin gene.

This report uses earlier product-approval votes as a historical prior for two questions. How often has a committee voted in favour of a product with labels like these? How often has the FDA later approved it? The table holds 156 meetings from 4 November 2016 through July 2026 at which a committee voted on approval of a specific product.

On meetings that asked an approval or benefit-risk question, the better-fitting models put a product with AMT-130's labels at a chance of 0.90 to 0.96 of a favourable vote, and a chance of 0.89 to 0.95 of eventual FDA approval, when the meeting is not held by either of the two neurology committees defined in the method. If that vote is favourable, the chance of approval in those models is 0.88 to 0.92. If the vote is unfavourable, the chance of approval is 0.28 to 0.37. The range is across a few models. It is not one blended number.

The labels explain little of who receives a favourable vote. The strongest association in the table is between the committee's vote and the FDA's later action. The table has no column for the strength of the clinical evidence, for the use of an external control group, or for accelerated approval. A high historical rate is the rate among products the FDA has already chosen to bring to a committee.

Introduction

An FDA advisory committee is a panel of outside experts. The panel votes. The agency decides. For a new drug or biologic, the vote that matters in this project is the one that asks whether the product should be approved, or whether its benefits outweigh its risks. A favourable vote means more yes votes than no votes. Abstentions are recorded and are not counted as yes votes.

AMT-130 has not had that meeting. The date and the committee are unset. The practical question is what the last decade of similar meetings implies before the briefing book exists, and again after the vote exists. The answer is a prior, in the ordinary sense of a base rate from past cases. It is not a forecast that folds in the Huntington's data package.

The work had three parts. The first was a search for product-approval votes that the table had missed. The second was a lookup of every model factor that was still blank, unverified, or ambiguous. The third was a comparison of every combination of those factors, followed by a prediction from the few combinations that fit best. A value was changed only when a public source supported the change. A missing record was not treated as a rejection, and it was not treated as evidence that a designation was absent.

The meeting file, the search log, the factor log, and the model output sit beside this report. This document is the explanation of what was done and what the numbers mean.

Method

The meeting table

Each row is one advisory committee meeting and one product, from 4 November 2016 through 2026, where the committee took a vote on approval of that product. The vote recorded is the pivotal vote: the last question at that meeting, for that product, that asks about approval, licensure, or whether benefits outweigh risks. Questions about what to do before approval, about labeling, about a risk program, or about a dose are not used as the pivotal vote. If no approval question was asked, the table uses the first effectiveness question, and the row is flagged so that choice is visible.

The FDA's later action is recorded in two ways. The final action is the latest known decision, so an application that received a complete response letter and was approved years later counts as approved. A complete response letter is the FDA's letter saying the application is not approved in its current form. The one-year action asks whether the product was approved within 365 days of the meeting. Withdrawn and pending applications are not counted as approved. Pending and unverified outcomes are left out of the vote-by-outcome counts and are counted separately.

The comparisons that follow drop four kinds of rows. A meeting that was superseded by a later vote on the same product is dropped. A tie is dropped. One meeting on Lynparza, coded as favourable only inside a genetic subgroup, is dropped because it is not a plain yes or no. An outcome that is still pending or unverified is left out of the likelihood and is counted.

Before any model was fit, the analysis set was checked against the published four-cell counts of vote by outcome. The cells matched. Those counts, for the final action, are the ones below. Each column is one combination of the committee vote and the FDA's later action.

Group	Favourable, approved	Favourable, not approved	Unfavourable, approved	Unfavourable, not approved	Meetings in the count	Left out as unverified or pending
All pivotal votes	90	6	10	43	150	1
Approval or benefit-risk questions only	72	6	8	30	116	0
Rare disease, orphan designation for the indication	40	2	2	11	55	1
Rare disease and priority review	29	1	1	4	35	0
Cell and gene therapy	10	0	0	1	11	1

Votes in the full table of 156 rows come from FDA minutes or transcripts for 137 rows and from press reports for 19. Outcomes come from an FDA document or openFDA, the agency's public file of drug-application records, for 112 rows, from press reports for 38, and from a Securities and Exchange Commission filing or another company document for 6. Press-sourced rows are real records with a weaker source. They are not described here as FDA-verified.

Data sources

The meeting list began as FDA advisory committee notices in the Federal Register. Votes were taken from FDA minutes, voting results, and transcripts, with press and company filings used only where the minutes could not be recovered. Approval dates and review type for drugs came from openFDA. Orphan status came from the FDA

orphan-drug designation list kept by the Office of Orphan Products Development. A published count of meetings from 2016 to 2021, Daval and colleagues in JAMA Health Forum in 2023, was used as a completeness check. That paper publishes committee-year totals, not a list of meetings.

Two further searches were run for this report, because a second pass through the Federal Register would have repeated the original build.

The first search read FDA advisory committee calendars and the past-meeting pages of the human-drug and biologics committees, including archived pages, from 4 November 2016 through 4 November 2026. Each date and committee was compared with the meeting table, with the list of meetings seen and set aside, and with the list of meetings that still lacked a usable vote. That last list is empty. Extra attention went to the cellular and gene-therapy committee, the two neurology committees, joint meetings, and the years 2022 through 2026, which the Daval count does not cover.

The second search was the Securities and Exchange Commission's full-text filing system, EDGAR, for current reports that mention an advisory committee vote and a tally. A filing by itself was not enough to add a row. A row was added only when a public source gave the date, the committee, the product or application number, the wording of the pivotal question, and a yes and no tally, with a link.

The calendar comparison produced 271 dated items. Of those, 129 were already in the meeting table and 91 were already on the exclusion list. The rest were joint meetings already stored under the lead committee, a second day of a meeting already stored, pharmacy-compounding meetings, class or policy discussions, or pages that named no application and no tally. One product meeting was set aside on purpose: sulopenem, on 9 September 2024, had an application number, and the FDA minutes show that the committee was not asked to vote.

One row qualified and was added. Solithromycin, for community-acquired bacterial pneumonia, was reviewed by the Antimicrobial Drugs Advisory Committee on 4 November 2016 under new drug application 209006. The same vote covered the intravenous application 209007. The FDA transcript records the benefit-risk question as 7 yes and 6 no. The sponsor's annual report to the Securities and Exchange Commission records a complete response letter on 29 December 2016 and states that the review was a priority review because the product was a qualified infectious disease product. The orphan-drug list has no row for solithromycin. The indication is a common infection, so orphan status is no. The vote source is the FDA transcript. The outcome and the priority designation are from the company filing.

Daval and colleagues counted 164 meetings for 2016 to 2021. After the start date of this table, pharmacy-compounding meetings, and joint meetings filed under another committee, about 19 of those published counts remain unmatched. Daval does not list the meetings, so those 19 cannot be identified one by one. They were not filled by adding rows. The model does not wait on them.

Factors that were still open

Every cell that the model might use, and that was unverified, ambiguous, or blank, was looked up. A cell was changed only when the source named the designation for the submission the committee actually reviewed. The priority attached to an original application was not copied onto a supplement or a resubmission. A decision date was not used to infer priority review, because a short clock is not the same statement as a priority designation.

Six priority values were filled.

Product	Meeting	Application	New value	Source
Ryoncil	13 Aug 2020	125706	Priority	FDA review memo. The memo says the application was filed as a priority review on 30 March 2020, and that is the application the committee reviewed.
Quizartinib	14 May 2019	212166	Priority	Daiichi Sankyo release, 21 November 2018. Press.
Omburtamab	28 Oct 2022	761176	Priority	Y-mAbs release, 31 May 2022. Press.
Omecamtiv mecarbil	13 Dec 2022	216401	Standard	A secondary write-up of Cytokinetics' 4 February 2022 acceptance notice.
Midomafetamine	4 Jun 2024	215455	Priority	Lykos release. Press.
Darzalex Faspro, high-risk smoldering myeloma	20 May 2025	761145, supplement 029	Standard	openFDA tags that supplement as standard. FDA minutes identify the supplement the committee reviewed.

Forty-five priority cells stay unverified. They are mostly products that were not approved, biologics the openFDA drug file does not cover, and supplements the file does not tag. The cell and gene therapy rows still open include Carvykti and Abecma at their 2024 supplement meetings, NurOwn, and the resubmissions of deramiocel and Tudriqev. Each of those resubmissions had priority review on the original application. That fact was not copied forward.

Two orphan cells stay marked for review, which means the designation text does not settle the indication the committee discussed. Darzalex Faspro on 20 May 2025 was reviewed for high-risk smoldering multiple myeloma. The orphan designation on file is for multiple myeloma, and it does not name the smoldering indication. Tudriqev, on 30 July 2026, has no orphan-drug row under that name, under vusolimogene oderparepvec, or under RP1. The absence of a row was not coded as "not orphan." Both cells stay "review," and any model that uses orphan status leaves them out.

Meeting type needed no change. Every row is already an initial application or a supplement. The ambiguous Lemtrada meeting that had been flagged in earlier work falls outside this window.

What the model uses

Six facts are known before the meeting, or are fixed by the kind of meeting. Each is coded yes or no.

Orphan status means the product had an FDA orphan-drug designation for the indication reviewed at the meeting. Orphan designation is for a disease that affects fewer than 200,000 people in the United States. Rows still marked "review" drop out of models that use this factor, and they stay in models that do not.

Priority review means the FDA had granted priority review for the submission the committee reviewed, as against a standard review. Rows still unverified drop out of models that use this factor, and they stay in models that do not.

Cell and gene therapy means the meeting was held by the Cellular, Tissue, and Gene Therapies Advisory Committee, or the product was one of four cell therapies reviewed at another committee: Kymriah, Ryoncil, Carvykti, and Abecma. Rebyota, a fecal microbiota product, is not in this group.

The central nervous system factor means the meeting was held by the Peripheral and Central Nervous System Drugs Advisory Committee or by the Psychopharmacologic Drugs Advisory Committee. A neurologic disease reviewed by another committee, including a gene-therapy committee, is coded as not in this group. AMT-130 is a treatment for a brain disease, and the committee that will review it has not been named, so every model that

contains this factor is reported twice.

Center means the application sits at the Center for Biologics Evaluation and Research, as against the Center for Drug Evaluation and Research. AMT-130 is a biologic and is coded at the biologics center.

Initial application means a first license or approval application, as against a supplement to a product already on the market. AMT-130 is an initial biologics license application.

The wording of the vote is used as a filter, not as a seventh yes-or-no factor. The main results use only questions about approval, licensure, or benefit against risk. A second pass uses every pivotal vote, including questions that asked only about effectiveness.

The committee vote itself is used only in the second stage, when the outcome is FDA approval. Favourable means yes votes exceed no votes.

Calendar year was not part of the full grid. It was added, as a straight line centered on 2020, only to the models the grid had already kept, and it was evaluated at 2026. The year 2026 is inside the years the table already contains.

How the models were fit

Every subset of the six factors was fit twice: once for a favourable vote, and once for eventual FDA approval. A third screen added the vote as a factor in the approval model. With six yes-or-no factors there are 64 subsets in each screen, including the model with no factors.

The fit is logistic regression. The probability of a yes outcome is a smooth S-shaped function of a weighted sum of the factors. The weights were estimated with a penalty that pulls them toward zero. The penalty matches a normal prior with standard deviation 2.5 on the intercept and standard deviation 1 on each factor. The same model was also fit with no penalty. If that unpenalized fit failed to converge, or if any weight exceeded 8 in absolute value, the fit was marked as separated. A separated fit can look precise while the data in a small cell are all successes or all failures. Separated fits are reported in the screen and are not used as a probability.

Fit is summarized by Nagelkerke's R-squared. This is a pseudo R-squared for a yes-or-no outcome. Zero means the factors do no better than a single overall rate. One would mean a perfect fit. It is not the R-squared of a straight line through a cloud of points. McFadden's R-squared, a simpler ratio of log likelihoods, is stored beside it in the output file. Models are ranked by the Akaike information criterion, which charges a model for each extra factor. A factor that does not improve the fit enough raises the criterion, and the smaller model is preferred. The criterion is computed from the unpenalized fit when that fit is stable.

For each outcome the prediction uses the four stable models with the best criterion, plus the single stable model with the highest R-squared if the criterion had not already kept it. That is a cap of about five models per outcome.

Two intervals are reported, and they answer different questions. The bootstrap interval resamples the meetings 400 times, refits the penalized model, and takes the 2.5th and 97.5th percentiles of the predicted probability. The seed is fixed, so the interval can be rebuilt. The stratum interval ignores the regression and looks only at the meetings that match AMT-130 on the factors in that model. With k successes in n such meetings, the interval is a Jeffreys interval, which is the 2.5th and 97.5th percentiles of a beta distribution with parameters 0.5 plus k and 0.5 plus n minus k. When n is under about eight, that interval is the honest description, and the logistic probability is borrowing strength from meetings that match on only some of the labels.

The main probabilities use the final action, so a later approval counts. The same kept factor sets were refit on the one-year action as a sensitivity. The all-pivotal-vote screen is a second sensitivity. Neither sensitivity is folded into the headline range.

AMT-130 enters every model as orphan yes, priority review, cell and gene therapy, biologics center, and initial application. The neurology-committee factor is set to no and to yes whenever the model contains it.

Results

The validity check passed: the meetings used in the model reproduce the published vote-by-outcome counts. The main set is the 116 approval or benefit-risk questions that have a final outcome. Of those votes, 101 come from FDA minutes or transcripts and 15 from press reports. Of those outcomes, 83 come from an FDA document or openFDA, 27 from press reports, and 6 from a company or Securities and Exchange Commission filing. Forty-five priority cells in the whole table remain unverified, so models that use priority are fit on 85 of these 116 meetings. The two orphan cells still marked "review" drop out of models that use orphan status. Among the 68 approval-question meetings that have both a final action and a one-year action, the two labels disagree 13 times.

Chance of a favourable vote

Priority review is in every model the criterion kept. Adding orphan status, an initial application, or the cell-and-gene-therapy and biologics-center labels raises R-squared by a small amount and does not improve the criterion by enough to displace the smaller models. The highest R-squared is 0.16, from priority review together with cell and gene therapy, the biologics center, and an initial application. That is a small share of the variation in the vote.

Factors	Meetings	R-squared	Probability	Bootstrap 95%	Matching meetings (favourable)	Interval for that cell
Priority	85	0.12	0.90	0.83 to 0.95	43 (40)	0.93 (0.83 to 0.98)
Priority, initial application	85	0.14	0.92	0.86 to 0.96	32 (31)	0.96 (0.86 to 1.00)
Orphan, priority	84	0.13	0.92	0.84 to 0.98	25 (23)	0.91 (0.77 to 0.98)
Orphan, priority, initial application	84	0.16	0.93	0.87 to 0.98	20 (19)	0.94 (0.79 to 0.99)
Priority, cell and gene therapy, biologics center, initial application	85	0.16	0.96	0.92 to 0.98	5 (5)	0.96 (0.62 to 1.00)

The probabilities run from 0.90 to 0.96. The top of the range is the five-meeting cell in the last row, and all five were favourable. The Jeffreys interval on that cell runs from 0.62 to 1.00. The logistic interval is narrower because the model borrows strength from the larger priority-review pattern. For a cell this small, the wider interval is the one that describes the cell itself.

None of these five models contains the neurology-committee factor, so the probability is the same if the meeting is later assigned to one of those two committees. The same-label counts are also unchanged, because the factor is not in the model.

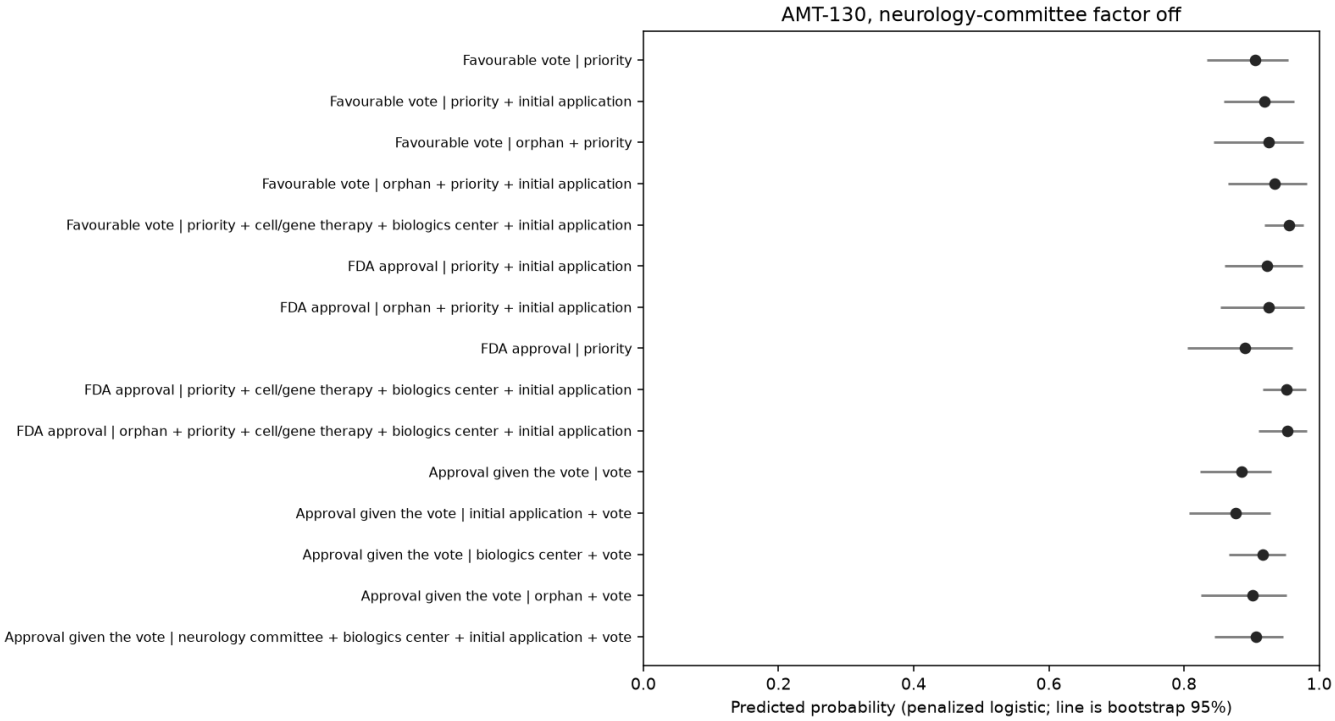


Figure 1. Predicted probabilities for AMT-130 when the meeting is not held by either neurology committee. Each point is the penalized logistic probability. The line is the bootstrap 95 percent interval.

Chance of FDA approval, before the vote is known

The same exercise for eventual approval, without using the vote, keeps a similar set of models. Priority review alone has an R-squared of 0.01. The predicted probability is still 0.89, because approval is already common in this table. The factor adds almost no discrimination. The highest R-squared among the kept models is 0.15.

Factors	Meetings	R-squared	Probability	Bootstrap 95%	Matching meetings (approved)	Interval for that cell
Priority, initial application	85	0.13	0.92	0.86 to 0.98	32 (30)	0.93 (0.81 to 0.99)
Orphan, priority, initial application	84	0.14	0.93	0.85 to 0.98	20 (19)	0.94 (0.79 to 0.99)
Priority	85	0.01	0.89	0.81 to 0.96	43 (38)	0.88 (0.76 to 0.95)
Priority, cell and gene therapy, biologics center, initial application	85	0.15	0.95	0.92 to 0.98	5 (5)	0.96 (0.62 to 1.00)
Orphan, priority, cell and gene therapy, biologics center, initial application	84	0.15	0.95	0.91 to 0.98	5 (5)	0.96 (0.62 to 1.00)

Across these models the probability of eventual approval runs from 0.89 to 0.95. The five-meeting cell is again all successes, and its own interval again runs from 0.62 to 1.00. Setting the neurology-committee factor to yes does not change these five probabilities, because none of the kept models includes that factor.

Chance of FDA approval, after the vote is known

Adding the vote changes the picture. R-squared rises to about 0.59. The criterion prefers a model that uses the vote and little else. Given a favourable vote, these models put the chance of eventual approval at 0.88 to 0.92. Given an unfavourable vote, they put it at 0.28 to 0.37.

Factors besides the vote	R-squared	If the vote is favourable	If the vote is unfavourable	Marginal probability	Bootstrap 95%, favourable case	Matching meetings if favourable (approved)
None	0.58	0.89	0.29	0.69	0.82 to 0.93	78 (72)
Initial application	0.59	0.88	0.28	0.67	0.81 to 0.93	58 (53)
Biologics center	0.59	0.92	0.37	0.82	0.87 to 0.95	9 (9)
Orphan	0.58	0.90	0.34	0.80	0.82 to 0.95	34 (32)
Neurology committee, biologics center, initial application	0.59	0.91	0.35	0.79	0.85 to 0.95	7 (7)

The marginal probability multiplies the model's own chance of a favourable vote by the chance of approval given a favourable vote, and adds the matching term for an unfavourable vote. It runs from 0.67 to 0.82. The low end comes from the first two rows, whose paired model of the vote does not include priority review or cell and gene therapy. That end is the historical base rate of a favourable vote, applied to AMT-130's approval equation. It is not the probability from AMT-130's full set of labels. The high end is the biologics-center model, whose matching cell is 9 approvals out of 9.

If the neurology-committee factor in the last row is switched on, the favourable-vote probability moves from 0.91 to 0.92 and the matching cell is empty. No meeting in the table was both a cell or gene therapy at the biologics center and a meeting of either neurology committee. That probability is an extrapolation.

Discussion

The headline range sits near 0.90 because the products in this table, and especially the products with priority review, have usually been approved after a favourable vote. Priority review alone barely moves the approval model: the R-squared is 0.01, and the probability of 0.89 is close to the overall rate of approval among meetings that have a known review type. The model is reporting a high base rate. It is not detecting a large effect of the label.

The vote is the factor that separates outcomes. An unfavourable vote is associated with a chance of eventual approval near one in three in the kept models. A favourable vote is associated with a chance near nine in ten. That gap is visible in the four-cell counts. Of 116 approval questions with a known final action, 72 were favourable and later approved, 6 were favourable and not approved, 8 were unfavourable and later approved, and 30 were unfavourable and not approved.

Two readings of "approved" diverge, and the divergence matters for a decision date. The final action counts a later approval. Ryoncil is the clearest example in this table: the committee voted in 2020, the FDA issued a complete response, and the product was approved in 2024. The final action is approved. The one-year action is not approved. Across the approval-question rows that have both labels, they disagree 13 times out of 68. Refitting the kept approval models on the one-year action lowers the probabilities to a range of 0.71 to 0.83 before the vote is known, and to about 0.68 to 0.79 given a favourable vote. For a target date such as 1 May 2027, the one-year reading is the closer analogue of the first decision. The final-action reading answers a different question, which is whether the product is approved eventually.

The cell and gene therapy cell that matches AMT-130 most closely has five meetings, all approved after a favourable vote, in the approval-question set. The logistic model pulls the probability to 0.95 or 0.96 and the bootstrap interval stays narrow. The interval computed from those five meetings alone runs from 0.62 to 1.00. Both numbers are in the tables so that the borrowing of strength is visible. A cell of five is a count. It is a weak basis for a percentage.

The committee assignment can still move a prediction, and it cannot move it with any matching history. Huntington's disease is a disease of the brain. In this table the neurology factor records the name of the committee, and the gene-therapy products were reviewed elsewhere. If the FDA seats AMT-130 at the gene-therapy committee, or at a cancer committee that has reviewed other cell therapies, the factor is off and the main tables apply. If the FDA seats it at the Peripheral and Central Nervous System Drugs Advisory Committee, the factor is on, and any model that needs a gene-therapy meeting of that committee is predicting into an empty cell.

A year term, evaluated at 2026, lowers the chance of a favourable vote into a range of 0.77 to 0.89 and improves the criterion for every kept vote model. Later meetings in this table have been somewhat less often favourable. The same term does not improve the approval models, and one of those year-augmented fits separates and is not used. The year 2026 is already represented in the table, through July 2026, so this is the fitted line at the right-hand end of the observed years.

What the model cannot see is the part a briefing book would be about. AMT-130's development has been discussed publicly in terms of an external control comparison and a 48-month result on the composite Unified Huntington's Disease Rating Scale, and the company has said the application is for accelerated approval. None of those facts is a column. Accelerated approval appears only as a feature of particular past votes, such as the Elevidys benefit-risk question, and it was not coded as its own factor. Two products with the same orphan, priority, and cell-therapy labels can have very different evidence. This prior treats them as exchangeable.

The source mix cuts the same way. Fifteen of the 116 approval-question votes, and 33 of the 116 outcomes, rest on press or company documents rather than an FDA record. The solithromycin row that this round added is an FDA transcript for the vote and a company annual report for the rejection and the priority designation. The historical rate is a rate in a table with that mix of sources.

Selection sits underneath the base rate. The FDA calls an advisory committee when the decision is uncertain enough to want one. Products that sail through do not appear. The 0.90 figure is a rate among products that reached a vote, with labels like AMT-130's. It is silent on products that never reached a committee.

Conclusions

For a product with AMT-130's regulatory labels, the meetings since November 2016 imply a chance of about 0.90 to 0.96 of a favourable approval vote, and about 0.89 to 0.95 of eventual FDA approval, if the committee is not one of the two neurology committees. Given a favourable vote, the kept models put eventual approval at 0.88 to 0.92. Given an unfavourable vote, they put it at 0.28 to 0.37.

The pre-meeting labels explain little. The best R-squared for the vote is 0.16. The best small model for approval, using priority review alone, has an R-squared of 0.01. The probability is high because approval after these meetings has usually happened, particularly after a favourable vote and particularly among priority-review products.

The one-year action is less optimistic, at 0.71 to 0.83 before the vote. That reading counts only an approval within a year of the meeting. The closest cell-and-gene-therapy cell has five meetings. Its own interval runs from 0.62 to 1.00, which is wide enough that the point estimate of 0.96 should not be quoted alone.

Orphan status is taken from FDA designation 525216. Priority review is taken from the company's November 2026 releases until an FDA letter confirms it. The clinical evidence, the external control, and the accelerated-approval request are outside this calculation.

Suggested improvements

Forty-five priority cells are still unverified. Each one should be closed from the approval letter, the review memo, or a company filing that names priority or standard review for the submission the committee reviewed. Carvykti, Abecma, NurOwn, and the deramiocel and Tudriqev resubmissions are the open cell-and-gene-therapy cases. A designation on the original application should stay off the resubmission unless the source says it carried over.

The two orphan cells marked "review" should be set to yes or no only if a source settles the indication. Darzalex Faspro for high-risk smoldering myeloma, and Tudriqev, do not meet that bar today.

A sourced column for accelerated approval, for a resubmission as against a first cycle, and for the use of an external control would address the largest hole in the factor list. Those columns would have to be built from primary documents, one row at a time, under the same rule used here: no inference from a missing record.

Once the FDA names the committee and the date, the unused branch of the neurology factor can be dropped, and the year term can be evaluated at that year. If the question of interest is the first action rather than eventual approval, the one-year models should be reported beside the final-action models, not only as an appendix.

Press-only tallies that are still thin should be checked against minutes where a tally exists. The project log already flags the Tzield count, and the Skysona and Zynteglo counts, as press or incompletely confirmed.

The residual of about 19 meetings against the Daval count is a difference in published totals. Closing it requires a meeting list, which that paper does not provide. Adding rows so that the totals match would invent meetings.

Where a matching cell has fewer than about eight meetings, the Jeffreys interval should be quoted with the logistic probability. The five-meeting cell in this report is the case that makes the practice necessary.

Appendix. Additional results

The tables in this appendix are sensitivities. They are not inside the ranges quoted in the summary. "Neurology committee on" means the meeting is assigned to the Peripheral and Central Nervous System Drugs Advisory Committee or the Psychopharmacologic Drugs Advisory Committee. Probabilities are from the penalized fit. R-squared is Nagelkerke's. The criterion is the Akaike information criterion, and a lower value is a better penalized fit.

A. All pivotal votes

This pass includes effectiveness questions as well as approval and benefit-risk questions. The chance of a favourable vote is a little lower than in the main set, at 0.85 to 0.95. The chance of eventual approval, before the vote, is 0.91 to 0.95.

Outcome	Factors	Neurology committee	Meetings	R-squared	Criterion	Probability
Favourable vote	Priority, initial application	Off	106	0.09	108.1	0.87
Favourable vote	Priority	Off	106	0.05	108.5	0.85
Favourable vote	Orphan, priority, initial application	Off	105	0.10	108.8	0.89
Favourable vote	Orphan, priority	Off	105	0.07	109.1	0.87
Favourable vote	Orphan, priority, cell and gene therapy, neurology committee, biologics center, initial application	Off	105	0.13	112.4	0.95
Favourable vote	Same six factors	On	105	0.13	112.4	0.93
Eventual approval	Priority, initial application	Off	106	0.15	79.5	0.91
Eventual approval	Orphan, priority, initial application	Off	105	0.16	80.9	0.91
Eventual approval	Priority, neurology committee, initial application	Off	106	0.16	81.0	0.91
Eventual approval	Priority, neurology committee, initial application	On	106	0.16	81.0	0.88
Eventual approval	Orphan, priority, neurology committee, initial application	Off	105	0.17	82.3	0.91
Eventual approval	Orphan, priority, neurology committee, initial application	On	105	0.17	82.3	0.88
Eventual approval	Priority, cell and gene therapy, neurology committee, biologics center, initial application	Off	106	0.18	84.0	0.95
Eventual approval	Same five factors	On	106	0.18	84.0	0.93

Given the vote, the all-pivotal approval models sit higher than the main set, because the wider set of questions has fewer discordant outcomes. The probability of approval given a favourable vote in the kept models runs from 0.94 to 0.97 with the neurology factor off.

Factors	Neurology committee	Meetings	R-squared	Criterion	Probability if favourable	Probability if unfavourable
Priority, initial application, vote	Off	106	0.64	45.9	0.95	0.60
Orphan, priority, initial application, vote	Off	105	0.64	47.9	0.95	0.60

Factors	Neurology committee	Meetings	R-squared	Criterion	Probability if favourable	Probability if unfavourable
Priority, neurology committee, initial application, vote	Off	106	0.64	47.9	0.95	0.61
Priority, neurology committee, initial application, vote	On	106	0.64	47.9	0.95	0.57
Priority, cell and gene therapy, biologics center, initial application, vote	Off	106	0.64	49.7	0.97	0.69
Cell and gene therapy, neurology committee, biologics center, initial application, vote	Off	149	0.65	107.8	0.94	0.36
Cell and gene therapy, neurology committee, biologics center, initial application, vote	On	149	0.65	107.8	0.94	0.38

The last row has an empty matching cell. It is the same extrapolation described in the results: a gene-therapy meeting at a neurology committee has not occurred in this table.

B. Approval within one year

These rows refit the models kept for eventual approval, using approval within 365 days of the meeting as the outcome. Meetings whose one-year action is unverified are left out. The probabilities fall relative to the final-action models.

Factors	Neurology committee	Meetings	R-squared	Criterion	Probability
Priority, initial application	Off	61	0.15	69.8	0.82
Orphan, priority, initial application	Off	60	0.11	71.0	0.83
Priority	Off	61	0.05	72.3	0.80
Priority, cell and gene therapy, biologics center, initial application	Off	61	0.16	72.9	0.71
Orphan, priority, cell and gene therapy, biologics center, initial application	Off	60	0.13	73.9	0.73

Switching the neurology factor on does not change these five probabilities. Given a favourable vote, the one-year models give the following.

Factors besides the vote	Neurology committee	Meetings	R-squared	Criterion	Probability
None	Off	68	0.32	73.1	0.76
Initial application	Off	68	0.34	74.1	0.76
Biologics center	Off	68	0.33	74.3	0.69
Orphan	Off	67	0.34	71.9	0.79
Neurology committee, biologics center, initial application	Off	68	0.36	76.6	0.68
Neurology committee, biologics center, initial application	On	68	0.36	76.6	0.77

The one-year models that include the vote have a lower R-squared than the final-action models that include the vote, about 0.32 to 0.36 as against about 0.59. Part of the association between a favourable vote and approval is an approval that arrives after the first year.

C. A term for calendar year

Year was added only to the models already kept, as a straight line centered at 2020, and the probability was evaluated at 2026. For the favourable vote, every kept model fits better with the year term, and the probability drops.

Factors	Criterion without year	Criterion with year	R-squared with year	Probability at 2026	Stable
Priority	74.1	72.7	0.18	0.77	Yes
Priority, initial application	74.7	73.9	0.19	0.81	Yes
Orphan, priority	74.8	72.6	0.21	0.80	Yes
Orphan, priority, initial application	75.4	73.6	0.23	0.83	Yes
Priority, cell and gene therapy, biologics center, initial application	77.6	76.6	0.21	0.89	Yes

For eventual approval the year term does not improve the criterion. One fit separates and is not used as a probability.

Factors	Criterion without year	Criterion with year	R-squared with year	Stable
Priority, initial application	53.7	55.6	0.13	Yes
Orphan, priority, initial application	55.4	57.2	0.14	Yes
Priority	56.5	58.0	0.03	Yes
Priority, cell and gene therapy, biologics center, initial application	56.9	60.0	0.12	No
Orphan, priority, cell and gene therapy, biologics center, initial application	58.6	60.4	0.16	Yes

D. Where the files are

The meeting table is `adcom_full_pivotal.csv` in this folder, 156 rows. The search log is `missing_row_search.csv`. The factor lookups are `factor_verification.csv`. The model output is `amt130_prediction.csv`, with the screen of every subset in the `amt130_screen_*.csv` files. The figure is `charts/amt130_prediction_forest.png`. The fit is `scripts/amt130_predict.py`, run from this folder. Corrections to the meeting table are made in `scripts/cur.py` and applied when the table is rebuilt. The table is not edited by hand.